

Voyage of Fate

Titanic Survival Analysis by Passenger Class & Gender

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1. The Problem

The RMS Titanic sank on April 15, 1912, with roughly 2,224 people aboard. This project sets aside popular myths about overloading and speed, and asks a narrower, checkable question using the Kaggle Titanic passenger dataset: how much did passenger class and gender actually shape who survived?

2. Data & Method

The raw dataset was cleaned — blanks removed, outliers handled, and the data remodeled for accuracy — leaving **714 complete passenger records** with usable class, gender, and survival fields. This sample is smaller than the ~2,224 people actually aboard; the two figures are reported separately throughout this report.

Sample composition:

Group	Count	Share
Male	453	63.4%
Female	261	36.6%
First Class	186	26.1%
Second Class	173	24.2%
Third Class	355	49.7%

3. Findings

3.1 Survival by Class

Class	Passengers	Survived	Rate
First	186	122	65.6%
Second	173	83	48.0%
Third	355	85	23.9%

3.2 Survival by Gender

Gender	Passengers	Survived	Rate
Female	261	197	75.5%
Male	453	93	20.5%

3.3 Survival by Class x Gender

Class	Female Surv.	Female Non-Surv.	Male Surv.	Male Non-Surv.
First	82	3	40	6
Second	68	6	15	8
Third	47	55	38	21

The gender gap held inside every class, but was not uniform. Third Class accounted for **85.9%** of all female non-survivors in the sample (55 of 64). A First Class woman had a **96.5%** survival rate; a Third Class woman's was **46.1%**.

3.4 Sample vs. Historical Record

Basis	Survival Rate
Sample (714 records)	40.6%
Historical (~2,224 aboard)	31.9%

4. Caveats & Limitations

- **Discrepancy found and corrected:** an earlier draft stated male survival at 28.8%. Recomputed directly from the class x gender breakdown ($93 \div 453$), the correct figure is **20.5%**, used throughout this report.
- **Sample vs. population:** all percentages except the 31.9% historical figure describe the 714-record sample, not all 2,224 people aboard.
- **Correlation, not causation:** class and gender are strongly associated with survival here; this analysis does not establish the mechanism (lifeboat proximity, deck location, evacuation order are plausible but unproven).
- **No age or fare data used** in this pass — natural next variables for extending the analysis.
- **Visualization 7** (male passengers onboard by class, percentage-only: 22.3% First, 21.9% Second, 55.8% Third) exists only as reported percentages in source materials; no chart was fabricated for it.

5. Conclusion

Passenger class and gender were both strongly associated with survival aboard the Titanic, and their effects compounded rather than operated independently. Being female mattered enormously, but being female *and* in Third Class still left less than a coin-flip's chance of survival (46.1%), while being female in First Class made survival close to certain (96.5%). Socio-economic status didn't just correlate with survival — it modulated how much the era's evacuation norms actually protected you.